

Detection and Recognition of Animals Using Yolo Algorithm

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Abstract—Animal recognition and tracking systems have the potential to provide accurate and cost-effective monitoring of animal populations and their behaviour in the wild, which can be crucial for conservation efforts and ecological research. This study explores the utilization of the YOLO version 3 object detection and recognition algorithm, in conjunction with Firebase technology, to trigger an application that sends real-time data to nearby safari vehicles for animal tracking purposes. However, it is important to consider the ethical and privacy concerns of both the animals and the people involved when developing and using animal recognition and tracking systems and to ensure that adequate training and support are provided for users, particularly in cases where it involves interaction with wild animals.

Keywords—Recognition of Animal, Yolo Algorithm, tracking systems, Firebase technology.

I. INTRODUCTION

Machine learning is a field of artificial intelligence that involves the development of algorithms and models that enable computers to learn from data and improve their performance on a task over time, without being explicitly programmed to do so. Machine learning algorithms can be used for a wide range of applications, including image and speech recognition, natural language processing, and predictive analytics, among others. The key idea behind machine learning is that computers can be trained to identify patterns in data and make predictions or decisions based on those patterns, without requiring human intervention or explicit programming. By training machine learning models on large data sets, programmers can create algorithms that can recognize patterns, make predictions, and take actions on their own. This can save human effort and time, as the machine can learn and improve its performance over time. The goal of machine learning is to create models that can make accurate decisions based on data, with little or no human intervention.

The YOLO V3 model is a popular deep learning algorithm that can perform on-the-go, real-time identification of objects. It is used to identify and track the movement of animals and humans in a forest area, which is important for safety and maintenance purposes.

The model works by training on a large data set of images and labels, allowing it to learn to recognize different objects and their characteristics. It uses a set of CNNs (Convolutional Neural Networks) to extract features from the pictures and

then makes predictions on the location and class of objects within the image.

In this project, the YOLO V3 model can be used to detect the presence of animals within a protected zone in the forest. When an animal is detected, the model can send the location and timestamp to the nearest tourist van, allowing them to take appropriate action and maintain a safe distance from the animal.

Overall, machine learning algorithms like YOLO V3 have the potential to improve efficiency and accuracy in object detection and recognition, making them an important tool across various industries, including digital cities, retail, and surveillance.

II. PREVIOUS STUDY AND RESEARCH IN THE AREA

Conducting a literature survey and researching different algorithms can help one gain a better understanding of the strengths and weaknesses of each approach and ultimately make an informed decision about which algorithm is best suited for the specific use case.

It's of great value to the proposed system that the YOLO algorithm is the most appropriate for animal detection in this project. YOLO is known for its speed and efficiency, as it can detect objects in real time while maintaining high accuracy. Its ability to detect small objects with good accuracy is a key advantage over co-existing detection algorithms for objects. Overall, YOLO is a popular and well-regarded algorithm for object detection, and it sounds like it will be a great choice for this animal detection project. Another recent development is the use of YOLO for detecting animals in drone imagery. This application is particularly useful for wildlife conservation and monitoring, as it allows for quick and accurate detection of animals in remote areas.

A paper [1] mentioned deals with using the benefits of machine learning and deep learning in detecting and classifying animals, particularly in the context of conservation efforts. It's good to know that CNN was found to provide reliable detection by focusing on its accuracy, while VGGnet seemed to be a better algorithm to use for the classification of animals to endangered species set. It's important to note that these algorithms were tested on still images, which may not fully capture the complexities of detecting and tracking animals in dynamic environments. Nonetheless, the findings of the paper can provide valuable insights for further research and development in this field.

In another paper [2] published in 2021, the experiment in this paper was held in a pre-controlled environment which does not highly match our real-life situations of red foxes in the wild. However, the study provides insights into the potential application of YOLO algorithm for animal behaviour analysis and monitoring in controlled environments.

Another paper [3] published in 2020 makes use of algorithms such as OpenCV and CNN surrounding animals living in forests. It is interesting to see the variety of approaches being used and the strengths and limitations of each algorithm. The use of novel datasets, such as the one collected manually from Google in the paper mentioned, can be particularly helpful for improving the accuracy and specificity of the model for a specific use case.

III. APPLICATION/USES IN THE REAL WORLD

The Animalia detection system has many potential applications in various industries and domains. It can play a vital role in wildlife conservation and management, as well as in monitoring and protecting crops and livestock. The system's ability to detect and track animals in real time can help prevent damage to crops, reduce the loss of livestock, and prevent animal human conflicts [4].

In addition to wildlife and agriculture, the Animalia detection system can be useful in security and surveillance applications. It can detect intruders, monitor borders, and prevent the smuggling of valuable commodities like sandalwood. The system's alerting mechanism can also be used in emergency response situations like stampedes, fires, or natural disasters to detect the number of people or animals trapped and send alerts to emergency responders.

Overall, the Animalia detection system can provide a powerful tool for improving safety and security in a variety of settings, from agricultural fields to national borders [5]. Its versatility and real-time capabilities make it a valuable asset for any industry or organization concerned with the detection and tracking of animals or people [6].

IV. PROPOSED METHODOLOGY

YOLO algorithm has been proven to be faster and more efficient than traditional CNN algorithms for object detection tasks. One of the main advantages of YOLO is that it processes the entire image only once, making it much faster than traditional object detection algorithms that perform multiple passes over an image. However, like any algorithm, YOLO has its limitations and trade-offs.

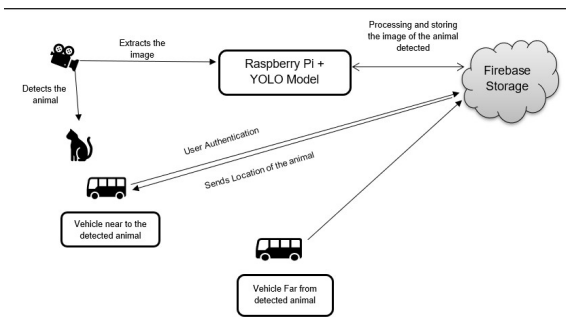


Fig. 1. Architectural diagram is a visual representation of the entire system's structure and components, showing how they interact and work together.

Figure 1 shows the interaction between the different sub systems in the software, where the images and videos captured using the camera are stored in the Firebase cloud storage via the raspberry pi. The User Details and the animal details captured are stored in the Firebase cloud. Once the user logs into the app, the details of which are stored in the cloud, the nearest animal detected is notified to the user through the app. Checking for the nearest animal is done by integrating google maps into our application. The User location is detected and sent to the Firebase cloud storage. Also, along with the images or videos, the location of each animal that is detected by the camera is sent to the Firebase storage so that the nearest animal detected is sent to the user having the application. Raspberry Pi is a powerful device that can run a full operating system and perform many of the same tasks as a desktop computer, such as word processing, web browsing, and even gaming. It can be connected to a monitor, keyboard, or mouse or controlled remotely over a network. Firebase Cloud Storage is mainly helpful in storing and retrieving data such as images, videos, and audio files from the cloud. This makes it easy to share and access data across multiple devices and apps.

A. You Look Only Once

YOLO (You Only Look Once) is an algorithm based on regression and detects objects by using only one neural network for predicting bounding boxes by using their class probabilities from their full pictures using a single run. This is in contrast to the pipeline-based systems used by many traditional CNNs for object detection, which involve multiple stages of processing. In YOLO [6], the input image is divided into a set of $S \times S$ grid cells, where each grid cell represents a specific region of the image. Each grid cell is associated with a fixed set of bounding boxes, and the network predicts the probability of each box containing an object, as well as the coordinates of the box relative to the grid cell. The equivalent vector mentioned for each grid cell in YOLO includes the following information:

The probability distribution across the different classes of objects that the network is trained to detect. By computing these values for each grid cell, YOLO can detect objects in the image with high accuracy and efficiency, making it a popular choice for real-time object detection applications.

- considering the object, we are inspecting it is there in the cell
- a label showing the representation of the subject or object
- minute presence of subject (by surrounded by a bounding box)

B. Non-Maximal Suppression

Non-maximal suppression is a step after processing commonly used in object detection algorithms, including YOLOv3, to eliminate overlapping bounding boxes and select the ones with the highest probability of containing an object [7].

The steps of Non-Maximal Suppression typically include:

- Sorting the bounding boxes based on their probability of the object being found scores (the probability that they contain an object).

- Choosing the box with the most probability of it being an object score and removing any other bounding boxes that have a high overlap with it (i.e., their intersection over union (Intersection over Union) value is above a certain threshold).
- Repeat the previous step with the remaining bounding boxes until no more boxes are left.
- The threshold for the Intersection over Union value is typically set to 0.5 or higher, depending on the specific application and the level of accuracy required.
- By using Non-maximal Suppression, redundant and overlapping bounding boxes are eliminated, resulting in a cleaner and more accurate detection output.

C. Intersection Over Union

This similarity or closeness between two bounding boxes is typically measured using the intersection over union (IoU) metric, which is a common technique employed by non-maximum suppression (NMS) to compare the overlap between two bounding boxes.

$$\text{IoU} = \frac{\text{Area of Intersection}}{\text{Area of Union}} \quad (1)$$

IoU [0,1] where 0 indicates no overlap and 1 indicates full overlap between the bounding boxes.

For example, if two bounding boxes are almost completely overlapping, their IoU value is close to 1. If two bounding boxes are completely separate and do not intersect at all, their IoU value is 0. If two bounding boxes are the same size and shape and are perfectly aligned, their IoU value is 1.

During NMS, a threshold value for the IoU is typically set to decide which bounding boxes to retain and which to discard. Bounding boxes with an IoU value above the threshold are considered redundant and are removed, while those with an IoU value below the threshold are retained.

D. Yolo version 2 and Yolo version 3

YOLOv3 is an upgraded version of YOLOv2 and it has several improvements over its predecessor. In addition to using the DarkNet-53 architecture, YOLOv3 also uses three scales for feature extraction instead of one, which allows it to detect objects at different scales more effectively. It also incorporates several other improvements such as feature pyramid networks, spatial attention modules, and more accurate bounding box predictions. All of these changes result in a more accurate and faster object detection system than YOLOv2.

YOLOv3 improves upon YOLOv2 and YOLO by implementing a feature pyramid network (FPN), which allows for feature maps to be extracted at different scales. By using FPN, YOLOv3 can detect objects of different sizes more accurately [8].

Additionally, YOLOv3 introduces skip connections, which allows for the fusion of higher-level features with lower-level features. This results in the preservation of fine-grained features, making it easier to detect smaller objects. By using a combination of FPN and skip connections, YOLOv3 can detect objects with high accuracy across a wide range of sizes, making it a popular choice for real-time object detection applications [9].

E. Mathematical Approach

The backbone network layer is responsible for extracting high-level features from the input image. The output of each convolutional layer is computed using the following equation:

$$Z = W * X + b \quad (2)$$

where Z is the output feature map, X is the input feature map, W is the filter weights, and b is the bias term.

Anchor boxes: YOLOv3 uses anchor boxes to predict the location and size of objects in the input image. Each anchor box is associated with a set of default aspect ratios and scales, and the predicted bounding box coordinates are computed using the following equations:

$$bx = \sigma(tx) + cx \quad (3)$$

$$by = \sigma(ty) + cy \quad (4)$$

$$bw = pw * e^{(tw)} \quad (5)$$

$$bh = ph * e^{(th)} \quad (6)$$

where bx and by are the centre coordinates of the predicted bounding box, bw and bh are the width and height of the box, tx, ty, tw, and th are the offsets learned by the network, and cx, cy, pw, and ph are the parameters associated with the anchor box. These equations allow YOLOv3 to predict the location and size of objects in the input image with high accuracy.

Loss function: The loss function used in YOLOv3 is a combination of localization loss, confidence loss, and classification loss. The total loss is computed using the following equation:

$$L = \lambda_{\text{coord}} * L_{\text{coord}} + \lambda_{\text{noobj}} * L_{\text{noobj}} + L_{\text{class}} \quad (7)$$

where Lcoord is the localization loss, Lnoobj is the confidence loss, Lclass is the classification loss, and lambdacoord and lambdanoobj are hyperparameters that control the weight of the different loss terms. The loss function is used during training to optimize the weights and biases of the network.

The Haversine formula is used to calculate the distance between two points on the earth that is the animal and the app user.

V. SALIENT FEATURES AND CHARACTERISTICS

Characteristics of this project mainly is to do with the capacities including the behaviours of the proposed system with the components.

A. Characteristics

A non-functional requirement of our project or product refers to the qualities and characteristics of the system that are not related to its specific functionality. These requirements describe how well the system performs its tasks, rather than what tasks it performs.

Some of the non-functional requirements of our animal and human detection system may include:

- Performance: The software system will be capable to detect animals and humans accurately and in real time. Reliability: The system will be reliable and able to detect animals and humans consistently, without any false positives or false negatives.

- **Scalability:** The system will have the capability to process a large volume of data, including images and videos, without any performance issues.
- **Usability:** The system will be friendly to use and understand, with an intuitive user interface and clear instructions.
- **Maintainability:** The system will be easy to maintain and update, with clear documentation and a modular architecture.
- **Compatibility:** The system will be able to work with different devices and operating systems.

Robustness: The system will be able to handle different lighting conditions, weather conditions, and other environmental factors that may affect the accuracy of the detection.

The main components/modules are:

- **Webcam:** This is the hardware component that captures the images of the animals that our algorithm will analyze. **Front-end application:** This is the user interface that is developed using Android Studio code. It will provide a way for users to interact with the application, such as by selecting the camera to use, viewing the recognized animals, and accessing other features.
- **Back-end algorithm implementation:** This is the heart of the application, where the image recognition algorithm is implemented. We have used pre-trained machine learning models or libraries to analyze the images captured by the webcam and identify the animals.
- **Firebase database:** This is the database used to store information about the users of the application and the identification code of the camera which detects animals. This data is used to improve the accuracy of the algorithm over time.
- **API calls:** API calls are made to various APIs to access the features and services that the application requires.
- **The front end of our software consists of a landing page with a login page, which then leads to a home page. This runs in the safari van.**

The pre-processing steps involve normalizing the pixel values, resizing the image to a fixed size, and reshaping the image to a tensor with the required shape. These steps help ensure that the input to the model is consistent and improves the accuracy and stability of the model's predictions. The first step of normalizing the pixel values of the image to range from 0 to 1 is important because it allows the neural network to easily learn patterns and relationships between the pixels in the image. Dividing all the pixel values by the maximum possible value (typically 255 for 8-bit images) is a common way to achieve this normalization.

The second step of resizing the image to a fixed size is also important because it allows the neural network to process images of a consistent size. This is necessary because most neural networks require inputs of a fixed size, and processing images of different sizes can cause problems such as memory overflow or underutilization. The third step of reshaping the image to a tensor with the required shape (typically (416, 416,

3) or (3, 416, 416)) is necessary because neural networks require inputs to be represented as tensors. The specific shape of the tensor depends on the data format required by the model being used.

For the algorithmic approach, OpenCV (cv2) is a very useful library for computer vision tasks, and it can be used for a wide range of applications including object detection, tracking, and image processing.

For detecting entry and exit in the application, we will use the cv2 library to capture and analyze images from the video feed. The `calcOpticalflow()` function in cv2 can be used for tracking motion between frames, which could be useful for detecting when an animal enters or exits the frame. The `ravel()` function can be used to reshape arrays, which could be useful for preprocessing the image data before feeding it into the neural network.

B. Functions

The cv2 library provides a function called VideoCapture that can be used to capture video from a built-in camera or external camera. The captured video can then be processed to obtain the necessary frames for object detection. In addition, it is important to preprocess the frames to be in the correct format and size for feeding into the YOLO algorithm. This may include resizing the frames to the appropriate size, converting them to the RGB format, and normalizing the pixel values. The preprocessing steps can be done using the cv2 library functions. Once the frames are preprocessed, they can be passed into the YOLO algorithm for object detection and classification. Overall, the cv2 library is a useful tool for capturing and processing video for computer vision applications such as object detection [10].

The Firebase database contains all the data of the users of the application and the results from the detection of the YOLO algorithm [11]. It provides real-time updates and synchronization, which means that any changes made to the database will be immediately reflected across all devices and applications that are connected to it. This can be useful for tracking the movement of animals in real-time, and for generating alerts or notifications when certain animals are detected.

Detecting and recognizing the subject animal of concern and notifying a safari van by making use of location proximity take the centre stage as it is the backbone of this system.

VI. AREAS OF IMPROVEMENT

A. Challenges in Our Existing System

- 1) Correct placement of the camera in outdoor environments where there may be complex backgrounds and occlusions is crucial.

The angle of the camera should be adjusted to minimize the overlap of objects and provide a clear view of the objects to be detected. In addition, the altitude of the camera should be considered as it can affect the depth and height of subjects or animals in the view range of the camera i.e., the field of view.

- 2) Non-max suppression [12] is a technique used in YOLO to eliminate duplicate detections by selecting the detection with the highest confidence score and suppressing all other overlapping detections. However,

Moreover, the problem of missing detections or false negatives can occur due to factors such as low lighting, occlusions, or quick movements.

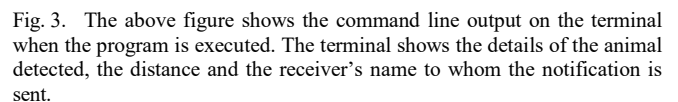
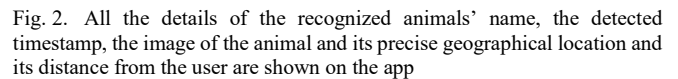
B. Future Scope of Improvement

- 1) As part of addressing the above-mentioned issues, techniques such as data augmentation, better training data, and fine-tuning the YOLO model can be used. Additionally, other algorithms such as the Kalman filter or optical flow can be used in combination with YOLO for better tracking and detection of objects. State-of-the-art capturing technology such as infrared capabilities or low-light sensitivity ensures further more accurate classifications, especially for night detection.
- 2) In areas where internet connectivity is not reliable, alternative solutions such as satellite internet or local network connectivity may need to be considered [14]. Additionally, it may be a good extension to set up local data storage and processing facilities to avoid relying entirely on internet connectivity for data transfer and analysis [15].
- 3) The software can be extended to include a machine learning algorithm to train the model on the behaviour of the animals thereby increasing the features, for example, to detect sick animals based on the regular actions of the subjects.
- 4) Tracking animals over time is an essential component of many wildlife monitoring and conservation efforts. Future research could focus on developing YOLO-based tracking algorithms that can track animals in real time across multiple frames, enabling researchers to monitor animal behaviour and population dynamics.

The front-end application is developed using Java in Android Studio, and the final output is displayed after the successful authentication of the user. Python has been used to implement the back end consisting of all the API calls and the algorithms to enhance maintainability and extensibility. The user information and the animal's identity are stored in the Firebase database. The IP address of the camera is fetched using protocols and IP scanners, and the coordinates of the same are forwarded to the Raspberry Pi OS. The coordinates are then transmitted to the van/driver, who can move to the location and track the animal. Once the integration of the interface with the algorithms was completed the extending of the system to Raspbian OS was done.

in real-time. The Go to Location button can be used to navigate to the location where the animal was detected using GPS or other location-tracking tools.

The forest application has a simple and user-friendly interface, which allows users to easily access information about animal sightings. In comparison to other existing systems, our system provides more accurate and precise information.



In conclusion, this paper has demonstrated the potential of the YOLO algorithm for animal detection in the forest regions, which can be an effective tool for mitigating human-wildlife interactions. The results showed that the system could accurately detect and recognize animals in real-time, providing relevant information like the name of the animal, timestamp, image, precise geographical location, and distance from the user. However, there are still some limitations and challenges, such as unreliable boundaries and proper camera placement

and the requirement for internet connectivity in the forest regions.

Future work should focus on addressing these limitations and expanding the capabilities of the system to include more species of animals and different habitats. It is hoped that this research will inspire more development and implementation

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